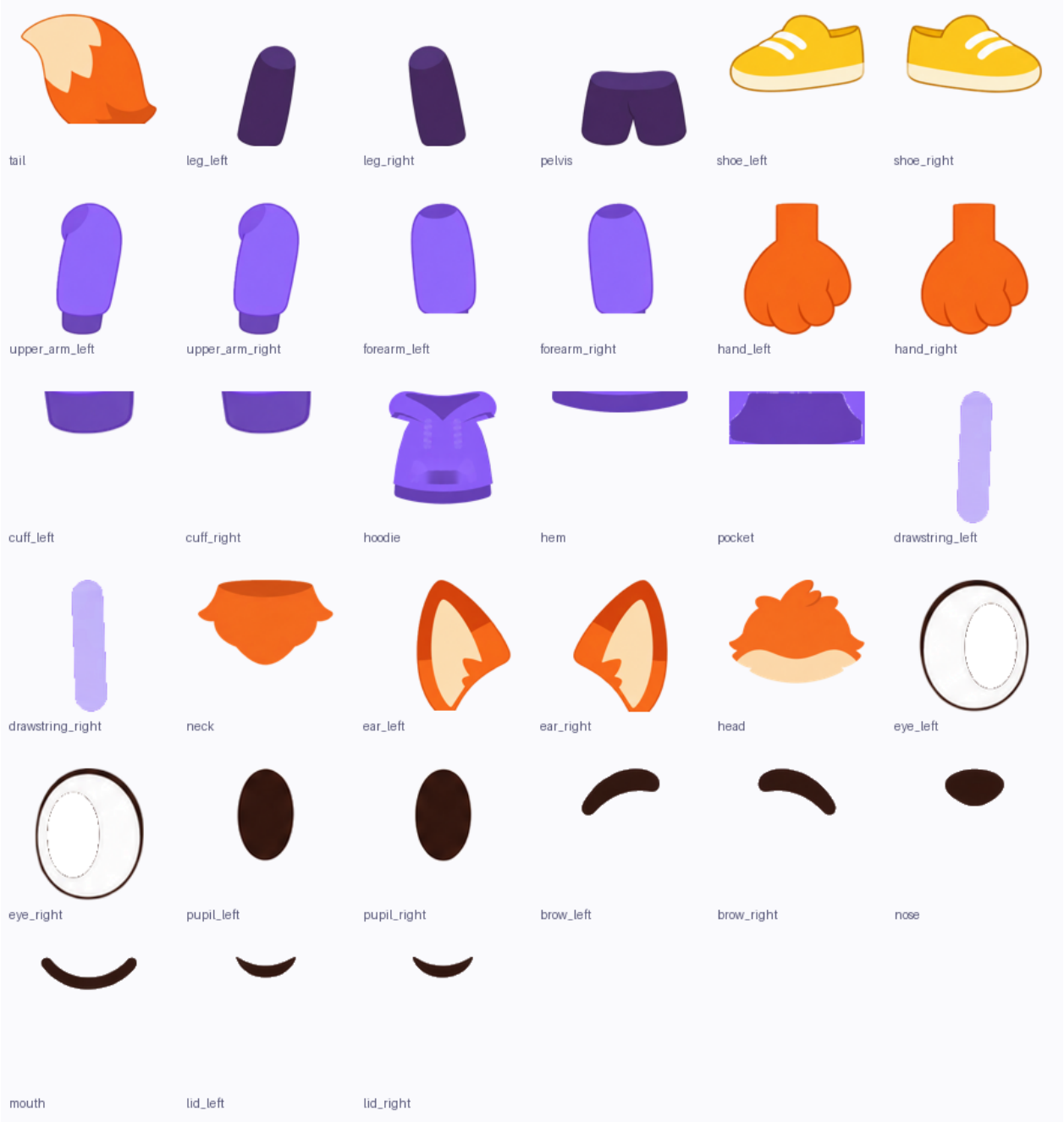
Clothing ownership, secondary motion and expressive rigs

Finding: a body-part label is not enough. Each visible garment detail needs an owner, an attachment and a deformation rule. Sleeves belong to the arm chain; cuffs follow the forearm/wrist; drawstrings are anchored to the hoodie. Wrinkle texture follows its garment surface.

This study extends the earlier mathematical-image report with an implemented 33-control Fox rig, a local timeline studio and measured tests. It is a project technical report, not a newly peer-reviewed paper. The research below informs engineering decisions; no universal emotional-accuracy claim is made.



Actual isolated assets and controls. Alternative palms and mouths are replacement assets. The original generated atlases and ownership masks remain saved.

Garment rules and deterministic mathematics

Separate motion responsibility from texture detail

The corrected torso mask excludes left and right arm fabric. Inpainting fills only the hidden cloth beneath extracted pocket and string artwork. The shoulder sleeve, forearm sleeve, cuff and hand form a hierarchical chain. Hood/chest wrinkle lines remain on the torso texture and deform with it; they are not treated as floating physical objects.

```
F_child = F_parent T(attachment) R(angle) S(scale)
image_world = F_child T(-local_pivot) image_local
```

Children inherit the parent joint frame, not a buffer offset. Part resizing preserves the world joint and scales its local pivot, child attachments and contact markers in the same asset-space units. Resizing and restoring are tested as an inverse pair. Viewport fitting happens later and does not change the anatomy.

Surface-attached details

```
a_deformed = deform(parent_mesh, a_bind)
a_child = a_child_bind + (a_deformed - a_bind)
```

Pocket, hem and drawstring anchors follow the local torso deformation. Current pocket/hem artwork remains a separate rigid detail whose anchor follows that surface; the implementation does not yet skin every detail pixel through a full cloth mesh. This distinction prevents overstating the rig as a complete cloth simulator.

Secondary angular response

```
theta'' + 2 zeta omega theta' + omega^2 theta
= -gain * driver_angle''
omega = 2 pi frequency
```

The drawstrings use a bounded damped response to the driving arm acceleration. The current preset uses 2.7 Hz and damping ratio 0.72; those are authored values, not experimentally optimal constants. The solver replays from zero at 1/240-second substeps with a final partial step. Output FPS and evaluation order do not change the state. A smooth terminal envelope settles the loop exactly.

Baraff and Witkin [1] study stable cloth integration; Small Steps [2] examines solver substeps. Our small angular system uses semi-implicit integration, not their full cloth/XPBD implementations. A later cloth solver would additionally need constraints, collisions, material calibration and validated skinning.

Emotion is a coordinated performance

Facial features, gaze, head angle, body posture, timing and secondary motion should support one intention. Independent controllability does not mean every part should move continuously. The current rig uses explicit mouth replacements, brow rotations, eye/pupil visibility, head/ear angles and authored arm tracks.

Intent	Primary cue	Supporting cue	Hold / settle
Hello	Arm raise and palm wave	Eye contact; light ear/tail response	Readable raised-arm hold; return
Thinking	Head tilt and asymmetric brows	Side gaze; concerned mouth	Pause before returning
Success	Both arms lift; open smile	Tail response; brief blink	Hold the achievement pose
Idle	Stable body and contacts	Small ear/tail action	Mostly still

Perceptual studies of facial-motion intensity [3,4] show why visual style, motion amplitude and perceived naturalness require evaluation. Their participants and characters differ from Fox. Increasing every motion amplitude is therefore not a justified rule for this mascot.

Triangular Character Animation Sampling [5] studies body motion, facial expression and relations together. It supports considering coordinated signals; our implementation uses authored presets rather than that learned energy model.

What still needs a perceptual experiment

Ask viewers to identify the intended action/emotion and rate readability, intensity and naturalness, without showing the preset name. Compare face-only, body-only and combined motion on held-out clips. Include confusing cases such as concern versus sadness. Report sample counts and disagreements. A technical score of 100 says nothing about recognition accuracy.

Technical defaults versus creative choices

Stable contacts, connected anatomy, finite transforms and non-flipped meshes are technical requirements. A six-degree head tilt, a particular smile or a two-second pause is an authored choice. Both belong in editable data, but only the former can be certified by geometry checks.

Render, measure, inspect, fix: actual results

Observed gap	Cause	Correction	Regression check
Sleeves stayed on torso	Generated hoodie included arm lobes	Source-pixel ownership masks; separate arm clothing	0 arm-mask pixels remain on torso
Block neck / clipped cheek	Poor overlap art and atlas boundary	Regenerated isolated clean plates	Visual review of bind and raised arm
Tail triangle lines	Anti-aliased clips left internal seams	Oversampled isolated mesh surface; overlapping clip coverage	Fully opaque synthetic interior
Scale could alter attachments	Image dimensions edited without bind-unit update	Pivot, children and contacts resize together	Resize/restore round trip
Fast success arm raise	Timing too short for chosen velocity target	Lengthened time; same trajectory	Time-normalized velocity
Editor or agent overwrites	Concurrent local revisions	Revision conflict checks + saved snapshots	MCP/user conflict test

Clip	Foot drift px	Joint gaps	Mesh flips	Connected body	Technical gate
hello	0	0	0	100.0%	Pass
thinking	0	0	0	100.0%	Pass
success	0	0	0	100.0%	Pass
idle	0	0	0	100.0%	Pass
joint-extremes	0	0	0	100.0%	Pass

The same time was evaluated at 24, 30 and 60 FPS with identical rig matrices. Browser checks cover playback, scrubbing, comparison, part editing, save, representation switches and a 390-pixel layout. Unit and MCP tests include rejected detached/off-canvas hands, bounded repair, JSON export and two-candidate generation.

Limitations: connected silhouette checks can miss internal shading or overlap errors; no calibrated emotion test has been run; the garment response is a small controlled model, not full cloth simulation. Learned semantic, depth and shadow adapters remain optional future work.

Research foundations and next validation

[1] Baraff and Witkin. Large Steps in Cloth Simulation. SIGGRAPH 1998.
Stable integration for stiff cloth systems. The QH drawstring oscillator is much smaller and does not implement their full method.
Primary paper / author source

[2] Macklin et al. Small Steps in Physics Simulation. SCA 2019.
Studies substeps and XPBD constraint behavior. It motivates testing solver time steps independently from output frame rate.
Primary paper / author source

[3] Hyde et al. Perceptual Effects of Damped and Exaggerated Facial Motion in Animated Characters.
Perceptual evaluation of rendering style and facial-motion amount. Applicability to Fox requires its own viewer study.
Primary paper / author source

[4] Assessing Naturalness and Emotional Intensity: A Perceptual Study of Animated Facial Motion. SAP 2014.
Studies expressiveness, recognition, intensity and naturalness. Supports testing those outcomes separately rather than using geometry as a proxy.
Primary paper / author source

[5] Triangular Character Animation Sampling with Motion, Emotion, and Relation. 2022.
A research framework coupling body, face and relational signals. Our current presets are authored, not generated by this model.
Primary paper / author source

[6] Igarashi et al. As-Rigid-As-Possible Shape Manipulation. SIGGRAPH 2005.
Local shape-preserving deformation is a future alternative to the current constrained texture mesh.
Primary paper / author source

[7] Lin et al. See-through: Single-image Layer Decomposition for Anime Characters. SIGGRAPH 2026.
Semantic decomposition and hidden-region completion remain separate from a finished garment rig. No learned layer model was installed here.
Primary paper / author source

Next improvements should be evidence-led

Test fabric extremes and additional character proportions; use surface-skinned details if the current rigid pocket becomes visibly wrong; validate the expression presets with viewers; then compare model-assisted decomposition against these authored masks. Keep previous accepted assets and candidates. A failed numerical gate requires a fix; an aesthetic disagreement needs an explicit visual decision.

Research and measurements checked 13 October 2026. This report records actual project tests, engineering synthesis and research applicability separately. Original full research is preserved in the earlier 23-reference report.